

Direct Interaction with a Reach-through Volumetric Display that uses an Elastic Diffuser

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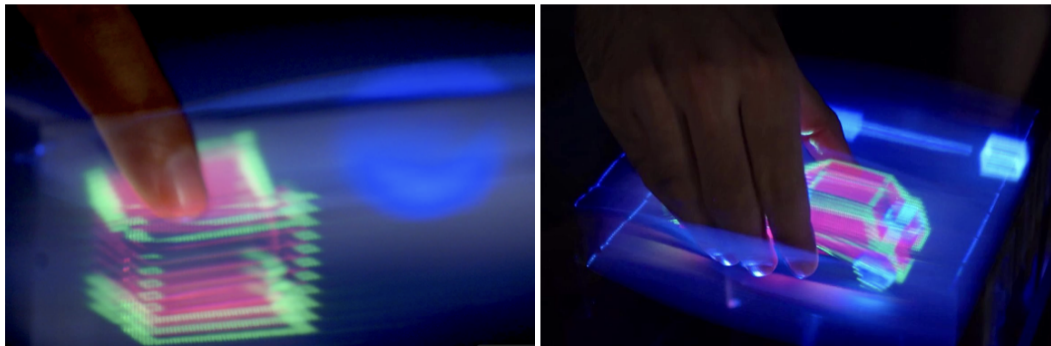


Figure 1: FlexiVol is a swept volumetric display that employs elastic strips as diffusers, enabling reach-through direct interaction with spatially aligned true-3D content: Left) touching a cube; Right) grabbing a car model.

Abstract

Volumetric displays render true 3D graphics without forcing users to wear headsets or glasses. However, the optical diffusers that volumetric displays employ are rigid and thus do not allow for direct interaction. We showcase FlexiVol, a volumetric display that employs elastic diffusers, allowing users to reach inside the display volume and have direct interaction with true-3D content. We plan to demo two playful come-and-interact applications involving FlexiVol, enabling participants to explore its interaction design space, without having to wear any devices. Participants will be able

to navigate, select, manipulate, or edit true-3D content, spatially aligned with their bare hands.

CCS Concepts

• **Hardware** → *Emerging interfaces*; • **Human-centred computing** → *Visualization systems and tools*; • **Computing methodologies** → *Graphics systems and interfaces*.

Keywords

Volumetric Display, True-3D, Direct Interaction, 3D Object Manipulation, Interaction Techniques, Swept Volumetric Display

ACM Reference Format:

Iosune Sarasate Azcona, Elodie Bouzbib, Unai Javier Fernández, Ivan Fernández, Manuel Lopez-Amo, Iñigo Ezcurdia, and Asier Marzo. 2026. Direct Interaction with a Reach-through Volumetric Display that uses an Elastic Diffuser. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799374>



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799374>

1 Introduction

We are used to seeing and interacting in a 3D world. Our vision uses cues to interpret depth, and our hands *directly* grab and manipulate the objects within. From the different 3D technologies, only volumetric displays and holograms provide all the depth cues [16]. Volumetric displays render true-3D and can provide most of the visual cues that we perceive from the real world. Nonetheless, with current technologies, users cannot reach inside the display volume to directly interact with virtual objects as they would do in the real world. Display diffusers are usually rigid, which causes safety hazards when reaching through; and interaction techniques are thus performed indirectly using a 3D mouse or a keyboard.

While several interaction techniques have been developed for 3D environments [5, 11], mostly indirect approaches have been adapted for volumetric displays [8] such as rays [7] or cursors. Other alternatives use hand gestures [2] or the index finger as a pointing device [9]. Commercial volumetric displays such as the Voxon VX1 [15] use traditional mouse and keyboard techniques, or 3D mice [1] as an input modality. Voxon displays also support a hand-tracking device (Leap Motion, Ultraleap Ltd.) enabling the use of the hands to manipulate virtual objects, but the real hands have to remain outside of the display volume. Methods based on touching the display boundary (i.e., protective case) [13] try to get closer to the display volume, yet the interaction remains indirect.

In our previous paper, “FlexiVol: a Volumetric Display with an Elastic Diffuser” [4], we investigated how to enable *reach-through interaction* to facilitate the selection and manipulation of true-3D content inspired from the definition of *direct interaction* from [10]: *allowing the user to literally get their hands into the virtual display and to directly interact with a spatially aligned 3D virtual world*. We proposed the use of elastic diffusers on swept volumetric displays to enable the user to reach inside the display volume. We drew inspiration from the use of elastic diffusers and their taxonomy of interaction techniques [18] to enable novel interaction techniques *within* a volumetric display with elastic diffusers.

We now propose to showcase FlexiVol, and enable participants to insert their hand within a volumetric display with an elastic optical diffuser and directly interact with spatially overlapped true-3D graphics. Using FlexiVol, the true-3D graphics and user hand provide coherent focus accommodation (Figure 1.Left), which allows for enhanced depth perception. The interaction with the rendered graphics is direct, the user can grab an object to move and rotate it (Figure 1.Right). We developed two mini-games, where users can explore freely and interact with all the elements within the scene; without having to follow a given scenario. These mini-games demonstrate FlexiVol’s interaction design space, previously discussed in FlexiVol’s paper [4].

2 Envisioned Interaction

In this section, we first depict FlexiVol’s interaction design space, and then propose two mini-games to showcase the interactions. The games can work with up to two hands including both handedness. Playing the two mini-games last less than 5 minutes. Mini-games were designed using Unity 2021.3.22.

2.1 Interactional Design Space

We first envisioned a set of interaction techniques to be used in reach-through direct interaction with volumetric displays. We selected the interaction techniques that are closer to real-life ones: techniques that use metaphors such as rays, clutching with the thumb or micro-gestures were discarded. We illustrate the interaction techniques as combinations of actions and gestures (Figure 2). FlexiVol’s interaction techniques are inspired from the literature involving touchscreens, multi-touch displays, and augmented/virtual reality 3D object manipulation [12, 17, 20].

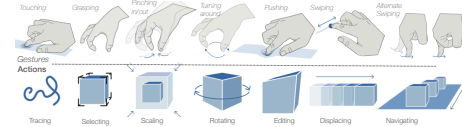


Figure 2: Reach-through direct interaction techniques for FlexiVol: Gestures and Actions

In Figure 2, a common mapping from *Gestures* to *Actions* is shown. However, other mappings are possible. For instance, *Selecting* could be performed by *Touching* or by *Grasping* between the fingers [6]. *Displacing* can be performed by *Pushing*, or by *Grasping*, and *Pinching*. We also depict a walking *Gesture* for *Navigating*: *Alternate Swiping* where users can move their fingers to mimic a mini-character walking inside the display.



Figure 3: We propose two mini-games to showcase our interaction design space. A. The village: (up) The user is free to explore a village, that hides a multitude of virtual elements to interact with. (bottom) The user starts a bonfire by selecting the wood. B. Come Fishy Fishy! The user can reach inside a fish tank and attempt to grab fish.

2.2 The Village

As a representation of our interaction design space, we propose a first mini-game, consisting in walking through a village. The participant first sees a terrain, a plain with grass and trees. They then need to perform an **alternate swiping** gesture to navigate through it and discover its interaction opportunities. The terrain contains four main areas to explore: the town center (with houses, a church, and a cemetery), a mountain (where hot air balloons float around), a lake (where a campfire is set), and a plain (with cows and trees).

Navigation. The navigation follows the hand orientation when alternate swiping on top of the oscillating bands, and starts with a threshold of four index and middle finger switches. The participant can thus explore the four corners of the virtual environment. When getting closer to its limit, the environment duplicates from its other corner, making it wrap-around and enabling infinite exploration in any direction.

Touching. Touching elements triggers events: touching the church starts the bells ringing, while touching the hot-air balloons makes them pop and re-appear further away, and touching the logs at the campsite starts or stops a bonfire (see Figure 3 - The Village). If the participant touches the water, they will hear the sound of their fall into it.

Rotating. Other elements, such as the cows or the windmill can be rotated. The user first grasps the object, using their thumb and index and/or middle finger; and change their orientation.

Editing. While navigating through the plain, the participant can find different types of mushrooms: ceps, morels, etc. They can touch and deform them by pushing on them. Up to a point where they will pop and split into multiple small mushrooms that spread around, so the participant can grow more if they want.

2.3 Come Fishy Fishy!

We also propose a second mini-game, where the participant can interact inside a fish tank (see Figure 3 - Come Fishy Fishy!). Participants will see different fish randomly moving around; and will be able to put their hands inside. The fish will avoid them; and the closer they get to a fish, the faster they go away. However, if a participant manages to touch a fish, it will start an animation. It is when the participant stops moving their hands around, that the fish will come and interact with them.

3 System Description

In this section, we describe the four parts needed for demoing FlexiVol. Beyond the visual display, other components are required to enable come-and-interact scenarios with true-3D content (see Figure 4).

3.1 Swept Volumetric Display

3.1.1 Hardware.

Projector: A high frame rate is needed to render into the display volume: if 100 slices are projected at a frame rate of 30 Hz, the projector has to operate at 3000 Hz. Usually, DLP projectors with time-multiplexation for the color are employed. The projected images are reduced to binary patterns and thus the original frame rate of the projector can be multiplied by the number of color bitplanes. For example, the Voxon VX1 projects 192 slices at 15 Hz (2880 Hz) with a resolution of 980×720px. When operating at 120 Hz with 24-bit color depth, it projects binary patterns at 2880 Hz with a resolution of up to 1280×800px.

Actuators: the typical actuators for a planar oscillating SVD are voicecoils (e.g., Voxon VX1 [15] and Dan Maloney's [14]) either custom designed or from a modified speaker.

Mechanical amplifier: the actuators may not oscillate with sufficient amplitude for the diffuser to provide a large enough display

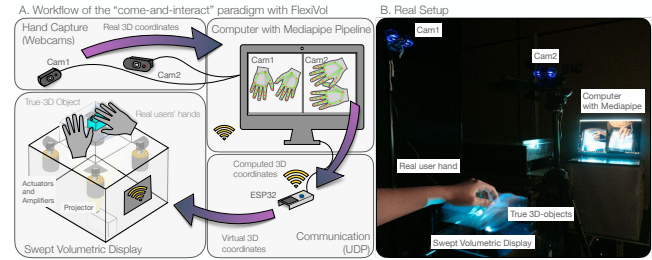


Figure 4: A. Workflow of the come-and-interact paradigm using FlexiVol: a user can come and interact with a true-3D object (turquoise) using their bare hands inside the swept volumetric display. Two webcams track the hand position, their video streams are processed and triangulated through Mediapipe. The computed coordinates are then transformed to the coordinate system of the volumetric display, and sent through UDP to the volumetric display - showing the users' hands 3D position in the volume as a 3D object, and enabling interactions with virtual content. B. Real set-up with a user hand being tracked.

volume. That is why some designs (e.g., Voxon VX1 and Dan Maloney's) use mechanical amplifiers. A mechanical amplifier can be just a spring or a lever mechanism.

Diffuser: it is usually a sheet of an optically diffusing material, which catches the projected light and diffuses it towards the viewers. The display volume is usually protected with an acrylic case to avoid users touching the solid diffuser. For FlexiVol, we replaced this solid diffuser with an array of elastic strips. We tried different materials: projection screens, elastane, latex and silicone; and found the elastic bands to present the best compromise of mechanical, optical and oscillatory properties.

3.1.2 Software.

Rendering: The virtual scene needs to be sliced along planes, those slices are projected at high-speed on the oscillating diffuser. Several slices are combined into a color image to be sent to the projector. For example, 24 slices with binary color are packed into a regular 24-bit RGB image that the projector expects from the HDMI or Display Port input.

Synchronization: In general, even if the actuators are excited with the same frequency as the projector, there will be drift and synchronization is required. If the actuator signal is used to synchronize the display, the rendering system has to sample the actuation signal before the start of each frame. That is, the rendering system checks the phase of the actuation signal and extrapolates the diffuser position for the next 24 slices, renders the slices and sends them packed as a color image to the projector. If the projector is the synchronization source, the vertical sync signal can be used to keep in a phased-locked-loop the actuation signal. The actuation signal is required to be a divider of the vsync signal. For example, in the Voxon VX1 the actuation signal is a 15 Hz sinusoidal and the vsync of the projector is a 120 Hz signal.

3.2 Hand Capture (Webcams)

To enable interactive scenarios where the user can actively perform interaction techniques, we need to capture and track its hand positions and movements. As the diffuser is oscillating at a high frequency, occlusion can occur, blocking the cameras views and hand capture. We thus chose to use **two** cameras (Papalook Web Camera HD 1080p); which we placed perpendicular to each other. One is oriented towards the volumetric display on the lateral side of the diffuser; while the other one is oriented towards its longitudinal side. They are placed higher than the display volume, to enable hand tracking above the volumetric display and a clear vision of the palm or wrist joints.

3.3 Computer with Mediapipe

The two cameras are connected to a computer; which runs a Python 3.8.20 terminal. We tried different hand tracking systems (Meta Oculus Quest, Leap motion), but chose a Mediapipe solution [19]. We triangulated hands points using the two cameras through Batpurev's real-time 3D hand pose estimation [3], and transformed the coordinates from cameras space to volumetric display space.

3.4 Communication (UDP)

Finally, the hand joint coordinates are sent through UDP (User Datagram Protocol). We programmed an ESP32 to work as a hotspot and UDP server, enabling the communication between the computer and the volumetric display. The computer sends "computed 3D coordinates" as packets of 63 values (x, y, z for 21 hand joints; 4 per finger + palm) in the new ordered basis to the server, which subsequently sends it to the display. These received coordinates are displayed onto the diffuser, giving the user a visual feedback of its hand 3D position in the volume. We allow for up to two hands to be present simultaneously within the volume.

4 Conclusion

We propose to showcase a modified swept volumetric display (FlexiVol) with an elastic diffuser to allow users to reach inside the display volume for interaction. This simple yet significant modification on volumetric displays opens up new opportunities for exploring the advantages of volumetric displays and direct reach-through interaction; we do so through two mini-games. Participants can navigate, select, rotate or edit true-3D elements following a come-and-interact approach, without having to wear any device (e.g., headset, controllers).

Acknowledgments

This research was funded by the European Research Council under grant agreement No 101042702 Intevol-ERC2021-STG.

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